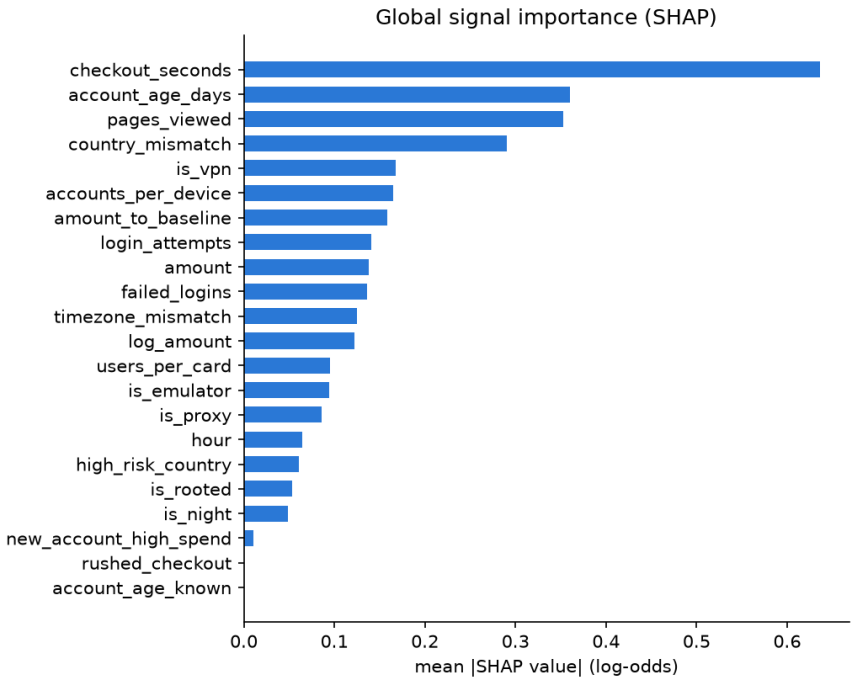
Explainability Report

Model gbm-20261110-0237-3afcd506 · generated November 09, 2026

How explanations work

Every order with an elevated score gets up to four plain-language reasons. For each signal that is genuinely abnormal on that order (e.g. a VPN is on, the card is linked to several accounts), the RiskScorer resets it to the value a typical legitimate order has and re-scores. The drop in the model's log-odds is that signal's impact. This is a local, model-faithful what-if method that needs no extra dependencies at serving time; SHAP and LIME are used offline to validate it.

Global importance



Top drivers overall: checkout_seconds, account_age_days, pages_viewed, country_mismatch, is_vpn.

Example: a blocked order

Reason shown to the merchant	Share of explained risk
Rushed checkout (7s from cart to payment)	41%
Device used by 20 different accounts	20%
Emulator / virtual device	20%
Device timezone doesn't match IP location	19%

Order T0019773: risk 82/100 → block.

Measured explanation quality

Over 181 correctly-caught fraud orders, **99%** of reasons cite a signal that is truly present (precision) and explanations cover **97%** of the main present signals.

Signal	Mentioned when present
abnormal_amount	91%
country_mismatch	78%
emulator	51%
few_pages	23%
high_risk_country	42%
login_failures	76%
new_account	20%
odd_hour	59%
proxy	73%
rooted_device	37%
rushed_checkout	98%
shared_card	56%
shared_device	56%
timezone_mismatch	66%
vpn	75%

Known weaknesses

- When many signals co-occur the model may lean on one dominant signal (checkout speed is the top feature) and under-cite others.
- Context signals (shared device/card) depend on how much history the store has.
- Reason text is template-based; the LLM fraud-analyst prompts (fra-*) test free-text explanations for jargon and clarity.